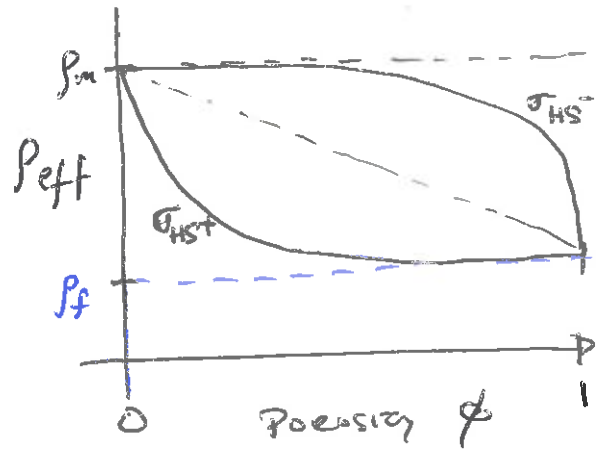
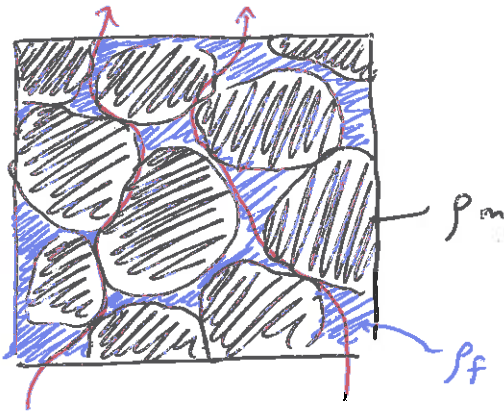


CONDUCTIVITY



SOLID PHASE CONDUCTION

$$\sigma_i = n_i q_i \mu_i$$

DEFECT " i " $\uparrow$ n_i (CONCENTRATION) $\uparrow$ q_i (CHARGE) $\uparrow$ μ_i (MOBILITY)

$$\mu_i = \mu_{0,i} \exp\left(-\frac{\Delta H_{\mu i}}{k_B T}\right)$$

MOBILITY μ_i $\leftarrow$ $\mu_{0,i}$ $\leftarrow$ $\Delta H_{\mu i}$ (ACTIVATION ENTHALPY)
 $\uparrow$ k_B (BOLTZMANN'S CONSTANT) $\uparrow$ T (TEMPERATURE IN [K])

$$n_i = N_i \exp\left(-\frac{\Delta H_{n_i}}{k_B T}\right)$$

$$\begin{aligned} \sigma_i &= N_i \exp\left(-\frac{\Delta H_{n_i}}{k_B T}\right) q_i \mu_{0,i} \exp\left(-\frac{\Delta H_{\mu i}}{k_B T}\right) \\ &= \underbrace{N_i q_i \mu_{0,i}}_{\sigma_{i,0} \text{ (PRE-MULTIPLIER)}} \exp\left(-\frac{(\Delta H_{n_i} + \Delta H_{\mu i})}{k_B T}\right) \end{aligned}$$

ΔH_i

$$\sigma_i = \sigma_{i,0} \exp\left(-\frac{\Delta H_i}{k_B T}\right)$$

(2)

$$\begin{aligned}\log \sigma_i &= \log \left(\sigma_{i,0} \exp \left(- \frac{\Delta H_i}{k_B T} \right) \right) \\ &= \log \sigma_{i,0} + \log \left(\exp \left(- \frac{\Delta H_i}{k_B T} \right) \right)\end{aligned}$$

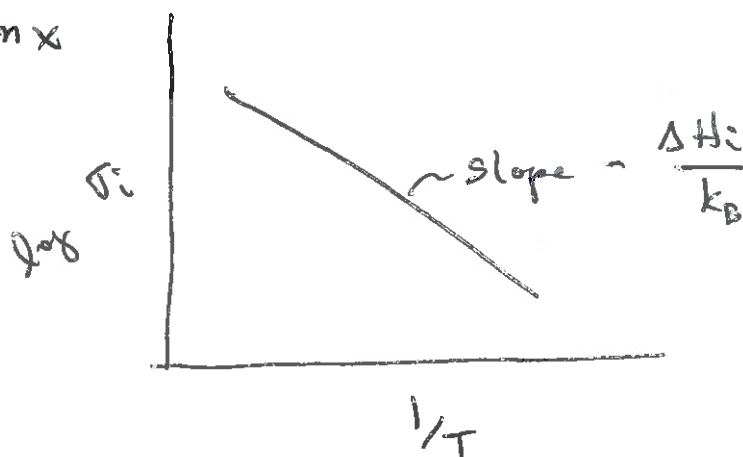
$$\boxed{\log \sigma_i = \log \sigma_{i,0} - \frac{\Delta H_i}{k_B T}}$$

$\log$ = natural $\log$

~~Answer~~ ARRHENIUS RELATIONSHIP

$$\log \sigma_i = y \quad \log \sigma_{i,0} = b \quad - \frac{\Delta H_i}{k_B} = m \quad \frac{1}{T} = x$$

$$y = b + mx$$



$$\sigma_{\text{eff}} = \sum_{i=1}^N \sigma_i$$

MAXWELL'S EQUATIONS

$$(1) \nabla \cdot \vec{E} = \frac{\rho_f}{\epsilon_0} \quad \text{GAUSS'S LAW ELECTRICITY}$$

$$(2) \nabla \cdot \vec{H} = 0 \quad \text{GAUSS'S LAW OF MAGNETISM}$$

$$(3) \nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad \text{FARADAY'S LAW}$$

$$(4) \nabla \times \vec{H} = \vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{AMPERE'S LAW}$$

$$(5) \vec{J} = \sigma \vec{E} \quad \text{CONSTITUTIVE RELATIONS}$$

$$5) \text{ into (4) } \nabla \times \vec{H} = \sigma \vec{E} + \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

$$3) \text{ into (4) } \nabla \times (\nabla \times \vec{E}) = \nabla \times \left(-\mu_0 \frac{\partial \vec{H}}{\partial t} \right) \quad \text{SWAP DERIVATIVE ORDER}$$

$$\nabla \times (\nabla \times \vec{E}) = -\mu_0 \frac{\partial}{\partial t} \left(\sigma \vec{E} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

VECTOR IDENTITY

$$\nabla \times \nabla \times \vec{A} = \nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$$

$$\text{From (1) } \nabla \cdot \vec{E} = \frac{\rho_f}{\epsilon_0} = 0$$

$$\nabla^2 \vec{E} = \mu_0 \sigma \frac{\partial \vec{E}}{\partial t} + \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

WAVE EQUATION

QUASISTATIC APPROXIMATION

$$\sigma \ll \omega \epsilon_0$$

↑
ANGULAR FREQUENCY

$$\nabla^2 \vec{E} = \mu \sigma \frac{\partial \vec{E}}{\partial t}$$

DIFFUSION EQUATION

(4)

$$\vec{E}(x, y, z, t) \xleftrightarrow{\sim} \tilde{E}(x, y, z, \omega)$$

time-domain frequency domain

$$1-D: \frac{\partial^2 \vec{E}}{\partial z^2} = \mu_0 \sigma \frac{\partial \vec{E}}{\partial t} \quad \text{Time-domain}$$

$$\frac{\partial^2 \tilde{E}}{\partial z^2} = -k_z^2 \tilde{E} \quad \text{Frequency-domain}$$

$$\frac{\partial e^{kz}}{\partial z} = k e^{kz}$$

$$\frac{\partial^2 e^{kz}}{\partial z^2} = k^2 e^{kz} \quad \text{CLOSE} \Rightarrow \text{NEED COMPLEX}$$

SOLN:

$$\tilde{E} = \underbrace{E^{(+)}(k_z, \omega) \exp(+i k_z z)}_{\text{UP GOING}} + \underbrace{E^{(-)}(k_z, \omega) \exp(-i k_z z)}_{\text{DOWN GOING}}$$

$$k_z = (1-i) \sqrt{\frac{\sigma \mu_0 \omega}{2}} \quad \text{WAVE NUMBER}$$

$$\uparrow$$

$$i \equiv \sqrt{-1}$$

REAL PART, SKIN DEPTH.

$$\delta = \left(\frac{2}{\sigma \mu_0 \omega} \right)^{1/2} = [m]$$

ELECTRICAL IMPEDANCE

(5)

$$\tilde{\mathbf{E}} = \mathbf{Z} \tilde{\mathbf{H}}$$

↑
IMPEDANCE

$$\mathbf{Z} = \frac{\tilde{E}_x}{\tilde{H}_y} = - \frac{\tilde{E}_y}{\tilde{H}_x} \quad \text{1-D.}$$

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} z_{xx} & z_{xy} \\ z_{yx} & z_{yy} \end{bmatrix} \begin{bmatrix} H_x \\ H_y \end{bmatrix}$$

$$E_x = \cancel{z_{xx}}^0 H_x + z_{xy} H_y$$

$$E_y = z_{yx} H_x + \cancel{z_{yy}}^0 H_y$$

IN 1-D, ~~dep~~

$$z_{xx} = 0 = z_{yy}$$

$$z_{xy} = -z_{yx}$$

IN 3-D

$$z_{xx} \neq 0, z_{yy} \neq 0$$

IN 2-D

$$z_{xx} = 0 = z_{yy}$$

$$z_{xy} \neq -z_{yx}$$

ID:

6

$$E_x = E_0 e^{-ik_z z}$$

$$H_y = H_0 e^{-ik_z z} = \frac{k_z}{\omega \mu_0} E_0 e^{-ik_z z}$$

$$Z_{xy} = \frac{E_x}{H_y} = \frac{E_0 \exp(\cancel{-ik_z z})}{H_0 \exp(\cancel{-ik_z z})}$$

$$= \frac{E_0}{H_0}$$

$$= \frac{E_0}{\frac{k_z}{\omega \mu_0} E_0}$$

$$Z_{xy} = \frac{\omega \mu_0}{k_z}$$

INPUTTING k_z

$$= (1+i) \left(\frac{\omega \mu_0}{2\sigma_a} \right)^{1/2}$$

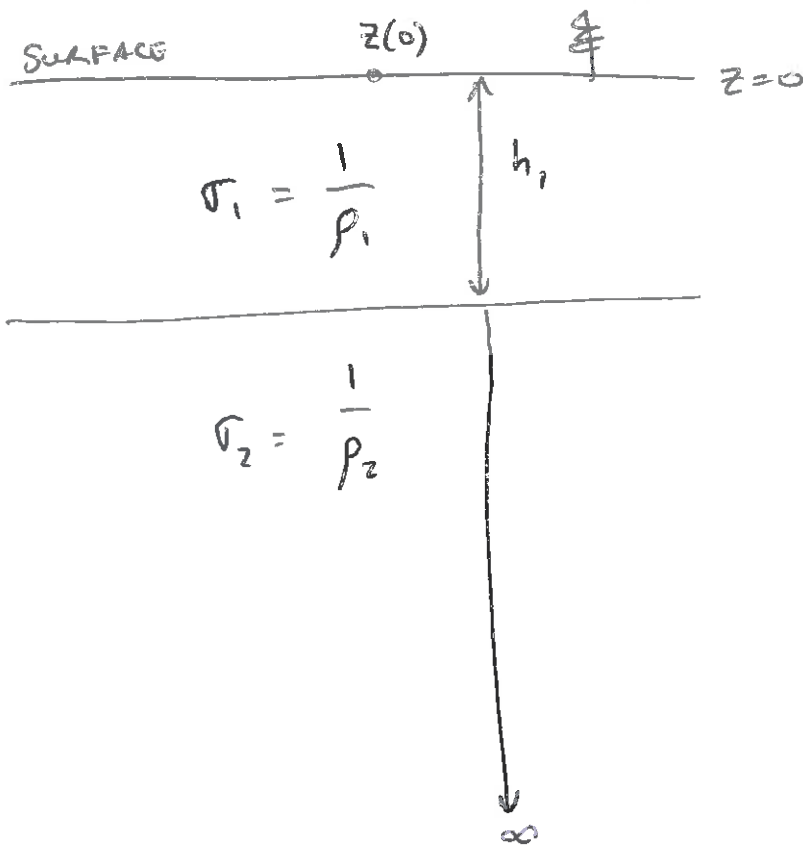
$$\sigma_a = \frac{1}{\rho_a}$$

$$Z_{xy} = (1+i) \left(\frac{\omega \mu_0 \rho_a}{2} \right)^{1/2}$$

$$\rho_a(\omega) = -\frac{1}{\omega \mu_0} Z_{xy}^2$$

$$\rho_a = \frac{1}{\omega \mu_0} |Z_{xy}|^2 \quad \text{APPARENT RESISTIVITY}$$

$$Z_{xy} = |Z_{xy}| e^{i\phi} \quad \phi - \text{PHASE}$$



$$Z(0) = \frac{\omega \mu_0}{k_1} \left(\frac{k_1 + k_2 \tanh(ik_1 h_1)}{k_2 + k_1 \tanh(ik_1 h_1)} \right), \quad \tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\tanh(0) = 0$$

$$\lim_{z \rightarrow \infty} \tanh(z) = 1$$

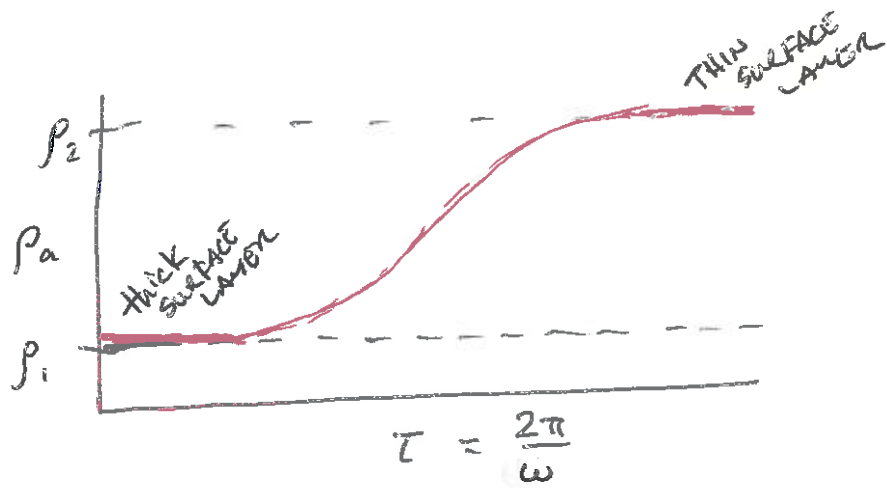
1. THICK SURFACE LAYER, $h_1 \rightarrow \infty$ OR $\omega \rightarrow \infty$

$$\lim_{h_1 \rightarrow \infty} \tanh(ik_1 h_1) \rightarrow 1$$

$$Z(0) = \frac{\omega \mu_0}{k_1} \left(\frac{k_1 + k_2}{k_2 + k_1} \right)$$

$$= \frac{\omega \mu}{k_1} = Z_1, \quad \rho_a \rightarrow \rho_i$$

(2)



2. THIN LAYER, $h_1 \rightarrow 0$ or $\omega \rightarrow 0$

$$Z(0) = \frac{\omega \mu_0}{k_1} \left(\frac{k_1 + k_2 \tanh(ik_1 z)}{k_2 + k_1 \tanh(ik_1 z)} \right)$$

$$\lim_{h_1 \rightarrow 0} \tanh(ik_1 h_1) \rightarrow 0$$

$$Z(0) = \frac{\omega \mu_0}{k_1} \left(\frac{k_1}{k_2} \right)$$

$$= \frac{\omega \mu_0}{k_2} = Z_2$$

$$\rho_a \rightarrow \rho_2$$

THIN-LAYER APPROXIMATION

$$Z(0) = \frac{\omega \mu_0}{k_1} \left(\frac{k_1 + k_2 \tanh(ik_1 h_1)}{k_2 + k_1 \tanh(ik_1 h_1)} \right)$$

$$\lambda = \frac{2\pi}{\omega} = [m] \quad h_1 = [m] \quad \lambda \gg h_1$$

$$|ik_1 h_1| \ll 1 \quad \tanh(ik_1 h_1) \approx ik_1 h_1$$

$$Z(0) \approx \omega \mu_0 \left(\frac{1 + ik_2 h_1}{ik_1^2 h_1 + k_2} \right)$$

(3)

3. HIGHLY CONDUCTIVE BASEMENT, $\sigma_2 \gg \sigma_1$, $\omega \rightarrow \infty$

$$\sigma_1 = \frac{1}{\rho_1}$$

$$\sigma_2 = \frac{1}{\rho_2}$$

$$k = (1-i) \left(\frac{\sigma \mu_0 \omega}{2} \right)^{1/2}$$

$$Z(\omega) \approx \omega \mu_0 \left(\frac{1 + i k_2 h_1}{k_2} \right)$$

$$\left[\begin{array}{|l|l|} \hline Z \approx i \omega \mu_0 h_1 & \rho_a \propto h_1^2 \\ \hline \end{array} \right]$$

SENSITIVE TO THICKNESS OF RESISTOR

4. HIGHLY RESISTIVE BASEMENT, $\rho_2 \rightarrow \infty$, $\sigma_1 \gg \sigma_2$

$$|k_1| \gg |k_2|$$

$$Z(\omega) \approx \frac{\omega \mu_0}{i k_1^2 h_1} \approx \frac{1}{\sigma_1 h_1}$$

$$\rho_a \propto \frac{1}{(\sigma_1 h_1)^2}$$

$\sigma_1 h_1 \equiv \text{CONDUCTANCE}$